B²SFormer: Hyperspectral Remote Sensing Image Classification Based on a Bilateral Branch Sparse Transformer Model

1st Leyuan Fan
School of Information Engineering
Ningxia University
Yinchuan, China
12023132014@stu.nxu.edu.cn

2nd Pengyuan Lv*
School of Information Engineering
Ningxia University
Yinchuan, China
lpydtc@126.com

3rd Jaijia Yu
School of Information Engineering
Ningxia University
Yinchuan, China
2896968121@qq.com

2025 IEEE 3rd International Conference on Computer, Vision and Intelligent Technology (ICCVIT) | 979-8-3315-7700-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCVIT67848.2025.11391382

***Abstract*—Hyperspectral images (HSIs) comprise numerous sequential spectral bands. This rich spectral data delivers detailed attribute information for material characterization. Recently, transformer models have been increasingly applied to HSI classification, owing to their capability to directly model long-range dependencies across spectral bands and achieve promising performance. However, the traditional transformer structure has difficulty in training on imbalanced HSIs because of the increased parameters of the model. In this paper, a bilateral branch sparse transformer (B²SFormer) framework is proposed to consider the class imbalance problem of hyperspectral data while emphasizing the key spectra in the classes with limited samples. In B²SFormer, sparse self-attention is introduced, which focuses on the features which contribute more, to better distinguish the different classes. A bilateral branch architecture is also proposed to learn a more robust network on imbalanced data. In the bilateral branch structure, normal and reverse sampling strategies are used for each branch, respectively. The first branch is used to model the original distribution of the data, and the second branch is used to improve the learning ability of the model on the imbalanced classes. The output features of the two branches are fused before the softmax layer to obtain the final prediction.**

Keywords-Hyperspectral image (HSI); Remote sensing; Classification; Imbalanced data; Transformer

I. INTRODUCTION

Hyperspectral remote sensing images (HSIs) provide a continuous spectrum, which is beneficial for identifying detailed types of materials [1]. In the early years, most HSI classification methods focused on exploiting the spectral signatures of pixels. The classical methods include clustering, statistical, and machine learning algorithms [2], [3].

Deep learning approaches have shown superior classification performance in HSI processing. Among the deep learning methods, convolutional neural network (CNN) models are the typical models, which can jointly extract deep spectral-spatial features by hierarchical convolutional kernels, and have achieved promising results in HSI classification. Lee et al. [4] introduced residual learning into HSI classification and investigated the impact of network depth on model performance. Mou et al. [5] has used the recurrent neural networks (RNNs) for HSI classification, introducing sequential modeling to the field.

The transformer model has provided a new perspective for modeling HSI data as it can be used for modeling the long-range spectral dependencies. The Transformer architecture, first introduced in natural language processing (NLP) [6], achieved remarkable success. Following this advancement, the Vision Transformer (ViT) was proposed to process images by modeling global information through multi-head self-attention (MHSA), subsequently inspiring its application in remote sensing. More recent work has explored Transformers for HSI classification, where studies have explicitly harnessed the self-attention mechanism to capture global dependencies among spectral bands [7] [8] [9].

The combination of CNN and transformer models has also been studied to combine the respective advantages of the two models [10].

In this article, we mainly analyze the performance of the transformer model in dealing with the class imbalance problem for HSI classification. The major contributions of this work are provided as follows:

1)A bilateral branch sparse transformer framework is proposed for HSI classification, which considers the class imbalance problem from the aspect of feature extraction and sample distribution. Furthermore, a shallow CNN layer is added at the front of the network to reduce the dimensionality of HSI data adaptively and improve the computational efficiency.

2)A sparse self-attention module is incorporated into the proposed framework to help focus on the key features in the tokens, which is beneficial for distinguishing the classes with limited samples.

3)A bilateral branch sparse transformer structure is proposed to solve the class imbalance problem, where an extra network branch is introduced based on a reverse sampling strategy to enhance the model's ability to capture the distribution of limited samples.

II. METHODOLOGY

A. Overview of the B²SFormer Structure

An overview of the proposed B²SFormer framework is displayed in Figure 1. The original HSI image is represented as $I \in \mathbb{R}^{h\times w\times c}$, where c is the number of the spectral bands and $h\times w$ is the spatial size. A sliding window is then applied on each pixel to split the original image into several patches $\in \mathbb{R}^{k\times k\times c}$, where k indicates the spatial size of the patch. The input patch $\mathbf{X}$ is sent into the bilateral branch structure to acquire the output class tokens $\mathbf{t}_{ran}$ and $\mathbf{t}_{rev}$. After this, $\mathbf{t}_{ran}$ and $\mathbf{t}_{rev}$ are fused and classified by the softmax classifier to acquire the final prediction for patch $\mathbf{X}$. Each branch in the proposed framework contains the same sparse transformer structure, which is introduced in the following part B. The parameters in each branch are learned from the training samples acquired by the random and reverse sampling strategies in the training step. In this way, the two branches can obtain different feature representations for the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:52:32 UTC from IEEE Xplore. Restrictions apply.

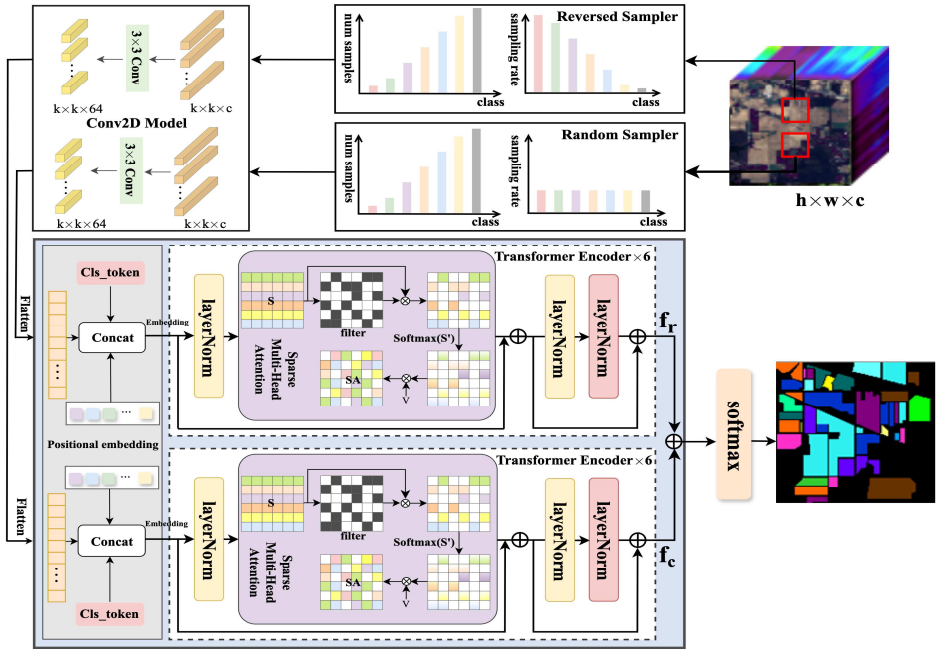


Figure 1. Overview framework of the bilateral branch sparse transformer framework.

samples. The training of the proposed method is introduced in part C.

B. Sparse Transformer

1) Shallow CNN Block

As described in part A, the input patch $\mathbf{X}$ contains the continuous spectral information. However, the abundant spectrum also introduces some redundancy and decreases the model efficiency. Thus, a shallow CNN block is designed to perform the task of initial feature extraction and reduce the data dimension. The structure of the shallow CNN can be described as follows:

$$\mathbf{X}' = \sigma(w\mathbf{X} + b) \tag{1}$$

where w represents the learnable weight matrix, b is the learnable bias, $\sigma(\cdot)$ is a rectified linear unit (ReLU) nonlinear function, and $\mathbf{X}' \in \mathbb{R}^{k' \times k' \times c'}$ is the output of the CNN block. c' is the number of channels after dimension reduction.

2) Transformer Encoder

The output $\mathbf{X}'$ of the CNN is then fed into the transformer encoder to acquire the global deep features. Firstly, $\mathbf{X}'$ is flattened into a token sequence $\mathbf{T} = \{\mathbf{t}_1, \mathbf{t}_2, ..., \mathbf{t}_m\}$, where m is equal to the spatial dimension of $\mathbf{X}'$ as $k' \times k'$. Each token $\mathbf{t}_i$ in the sequence is a c'-dimension vector. The benefit of the token sequence $\mathbf{T}$ is that both the spectral and spatial information of the input patch $\mathbf{X}$ are considered. Inspired by the ViT model , a class token $\mathbf{t}^{cls}$ is introduced into the patch token $\mathbf{T}$ to obtain a new token $\mathbf{T}^{cls} = \{\mathbf{t}^{cls}, \mathbf{t}_1, \mathbf{t}_2, ..., \mathbf{t}_m\}$, which helps to aggregate the global characteristics of the token sequence for the classification step. After this, the position code is embedded in $\mathbf{T}^{cls}$ to obtain $\mathbf{T}'$, which can retain the spatial location information in the input patch. The positional embedding can be represented as:

$$\mathbf{T}' = \{\mathbf{t}^{cls}, \mathbf{t}_1, \mathbf{t}_2, ..., \mathbf{t}_m\} + pos_embedding \tag{2}$$

where $pos_embedding \in \mathbb{R}^{m \times c'}$.

The MHSA mechanism is then performed on $\mathbf{T}'$ to consider the correlation among different tokens and channels. In the self-attention module, firstly, each token in $\mathbf{T}'$ is multiplied by the predefined learnable parameter matrix $\mathbf{W}_q$, $\mathbf{W}_k$, $\mathbf{W}_v$ to obtain $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$, which can be represented as follows:

$$\mathbf{Q} = \mathbf{W}_q \mathbf{T}' \tag{3}$$

$$\mathbf{K} = \mathbf{W}_k \mathbf{T}' \tag{4}$$

$$\mathbf{V} = \mathbf{W}_v \mathbf{T}' \tag{5}$$

We can obtain an attention score matrix $\mathbf{S}$ by multiplying $\mathbf{Q}$ and $\mathbf{K}$, which is shown as follows:

$$\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d}} \tag{6}$$

where d is a scale parameter.

$\mathbf{S}$ is then normalized by the softmax operation and multiplied by $\mathbf{V}$ to obtain the attention representation. The MHSA mechanism can be represented as:

$$SA = soft\max(\mathbf{S})\mathbf{V} = soft\max\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d}}\right)\mathbf{V} \tag{7}$$

$$MHSA = concat(SA_1, SA_2, ..., SA_h) \tag{8}$$

where h represents the number of heads. The $soft\max(\cdot)$ function is used to normalize the output. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:52:32 UTC from IEEE Xplore. Restrictions apply.

MHSA mechanism means that it is possible to capture different feature information at different ranges, and the combination of this information can effectively prevent overfitting.

3) Sparse Self-Attention Mechanism

In the traditional transformer encoder, the MHSA mechanism can capture the global context information and model the long-term dependencies. However, considering the global dependency can bring some small weights, indicating weak correlation between features. The increase of the number of small weights has the effect of decreasing the contribution of the major features for classifying the different classes. In the sparse self-attention module, a predefined weight threshold is used, and the weight values which are lower than the threshold are set to 0. Thus, the features with high correlation are emphasized. The details of the sparse self-attention operation are as follows:

$$\mathbf{S}_i' = Max_p(\mathbf{S}_i) \tag{9}$$

$$v_i = \min(\mathbf{S}_\mathbf{i}') \tag{10}$$

$$S_{i,j} = \begin{cases} S_{i,j}, S_{i,j} \geq V_i \\ 0, S_{i,j} < V_i \end{cases} \tag{11}$$

where i,j respectively represent the rows and columns of the attention score matrix $\mathbf{S}$ in (6). $\mathbf{S}'$ is a maximum matrix, and the value of each row in S comes from the first p larger values of each row in $\mathbf{S}$. V_i is the threshold, which is the minimum value of row i in S'. After the sparse self-attention operation, the MHSA mechanism is also performed based on (7) and (8) to acquire the output class tokens $\mathbf{t}_{ran}$ and $\mathbf{t}_{rev}$ for each branch.

C. Bilateral Branch Sampling of the Proposed Method

In the training step, the samples are generated by the random sampling and reverse sampling strategies from the training set for each branch. The random sampling strategy randomly selects the samples under the same probability to keep the original distribution characteristics of the data. For the reverse sampling strategy, the sampling probability of each sample is inversely proportional to the number of samples. The greater the number of samples in a class, the less the sampling possibility of this class. The reverse sampling probability can be described as follows:

$$W_i = \frac{N_{\max}}{N_i} \tag{12}$$

$$P_i = \frac{W_i}{\sum_{j=1}^{c} W_j} \tag{13}$$

where the number of samples for class $N_{\max}$ is N_i, and the maximum number of samples for all the classes is $N_{\max}$. c represents the number of classes. The sampling probability of each category is denoted as P_i.

The output tokens $\mathbf{t}_{ran}$ and $\mathbf{t}_{rev}$ are fused according to (14) and (15) to acquire the final class token before the softmax classifier.

$$f = \alpha \mathbf{t}_{ran} + (1-\alpha)\mathbf{t}_{rev} \tag{14}$$

$$\alpha = 1 - \left(\frac{T_c}{T_{\max}}\right)^2 \tag{15}$$

where α is a weight parameter, T_c and $T_{\max}$ respectively represent the current number of iterations and the total number of iterations, which decreases as the number of training epochs increases, ensuring that the two branches of different targets can show their own respective advantages without interfering with each other during the training process.

Thus, the loss function can be formulated as:

$$L = \alpha E(p, y_{ran}) + (1-\alpha) E(p, y_{rev}) \tag{16}$$

where E is the cross-entropy loss function, p represents the predicted vector of the model output, y_{ran} is label of the input sample for the random branch, and y_{rev} is label of the input sample for the reverse branch.

III. Experiments

A. Data Description

1) Indian Pines Dataset (IP)

The Indian Pines (IP) dataset features a 145 × 145 pixel image at 10-meter resolution, covering 16 land-cover categories, and utilizes 200 spectral bands following the removal of 20 absorbent and noisy bands.

2) Pavia University Dataset (PU)

The Pavia University (PU) dataset is a 610 × 340 pixel image featuring nine land-cover categories. From the initial spectral data, 12 noisy bands were discarded, leaving 103 bands for the experiment.

B. Experimental Settings and Evaluation Metrics

All experiments were conducted on the Python platform using an NVIDIA GeForce 3090 GPU. We employed the Adam optimizer with an initial learning rate of 5e-4, a batch size of 64, and trained the model for 500 epochs. A randomly selected subset comprising 10% of the training samples was used for training. Our evaluation was based on four key indicators, which encompassed Average Accuracy (AA), Overall Accuracy (OA), the Kappa coefficient (K × 100), along with the precision achieved by each class.

C. Parameter Analysis

This section analyzes two key parameters influencing classification performance: patch size and the number of Transformer encoders. Table I presents the effect of patch size (ranging from 5 to 11) on the two datasets. Furthermore, Table II compares the impact of the encoder under different training sample ratios on the Indian Pines dataset. The classification accuracy improves with an increase in the number of training samples.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:52:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I. THE EFFECT OF PATCH SIZE ON OVERALL AVERAGE ACCURACY ON THE TWO DATASETS

Patch size	Indian Pines(AA%)	Pavia University(AA%)
5 × 5	95.75	99.48
7 × 7	98.02	99.79
9 × 9	98.47	99.88
11 × 11	98.19	99.93

TABLE II. ABLATION STUDY ON THE NUMBER OF ENCODERS UNDER DIFFERENT TRAINING SAMPLES (INDIAN PINES DATASET)

Training set ratio	Encoder number	AA(%)
3%	3	91.33
	6	90.05
	9	89.81
	12	90.38
5%	3	96.03
	6	96.06
	9	95.88
	12	95.63
7%	3	97.36
	6	97.26
	9	96.90
	12	97.32
10%	3	98.52
	6	98.67
	9	98.54
	12	98.47

D. *Ablation Study*

To quantitatively assess the effectiveness of our proposed modules, a comprehensive set of ablation experiments was performed on the two datasets. The traditional self-attention mechanism based transformer with random sampling for training was used as the baseline method. The effectiveness of the sparse self-attention and reverse sampling strategies was analyzed. The results in Table III (rows 1-3) clearly demonstrate that the dual-branch transformer, which integrates both random and reverse sampling strategies, achieves better classification results than models relying on a single branch alone. This is because the reverse sampling strategy prefers to select the categories with fewer samples in the training set, which effectively solves the problem of category imbalance. As shown in the fourth row of Table III, the integration of both modules yields the highest classification accuracy. Furthermore, the computational efficiency of each method, measured by parameters and FLOPS on the Pavia University dataset, is provided in Table IV.

TABLE III. ABLATION EXPERIMENT RESULTS ON THE TWO DATASETS

	Reverse sampling branch	Sparse self-attention	Indian Pines		Pavia University	
			OA(%)	OA(%)	OA(%)	OA(%)
1	×	×	94.04	89.22	99.46	99.16
2	×	√	95.58	94.47	99.58	99.24
3	√	×	97.48	95.83	99.84	99.72
4	√	√	98.71	98.16	99.90	99.85

TABLE IV. COMPUTATIONAL COMPLEXITY AND MODEL SIZE ON THE PAVIA UNIVERSITY DATASET.

Methods	Parameters	Flops
1D-CNN	16.29M	1.046G
ResNet50	23.71M	0.034G
SF	615.41M	92.657K
SSFTT	234.39M	148.033K
CS2DT	1.70M	145.30M
GSCViT	78.66K	4.964M
B^2SFormer	264.96K	13.014M

E. *Comparison of Classification Results of Different Methods*

On the Indian Pines dataset (Table V, Figure 2), B^2SFormer shows competitive results. A more conclusive demonstration of its superiority is found on the Pavia University dataset. As quantified in Table VI, our method attains the highest accuracy. These quantitative findings are corroborated by the visual results in Figure 3, which reveal that B^2SFormer produces more precise classification maps, especially in heterogeneous regions and for tail categories with limited samples, validating its efficacy in capturing discriminative features.

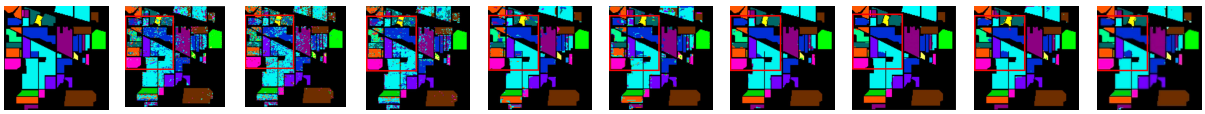
TABLE V. CLASSIFICATION RESULTS OF THE VARIOUS METHODS FOR THE INDIAN PINES DATASET

Class no.	SVM[3]	RNN[5]	1D-CNN[11]	ResNet-50[12]	SF[7]	SSFTT[10]	CS2DT[8]	GSC-ViT[9]	Proposed
1	50.00	48.78	74.35	92.68	81.39	87.80	**100.00**	94.59	**100.00**
2	77.17	66.77	80.10	92.45	83.00	95.25	**99.04**	97.20	98.44
3	67.11	45.52	52.76	86.07	92.32	98.20	98.15	98.80	**99.06**
4	46.84	31.46	50.24	79.81	65.07	95.77	**99.09**	89.47	97.65
5	86.13	85.29	87.22	93.33	94.37	98.16	99.06	97.15	**99.77**
6	95.62	91.02	97.26	97.41	93.50	99.54	98.66	98.29	**99.69**
7	89.29	36.00	86.95	68.00	73.07	99.93	98.61	95.45	**100.00**
8	96.03	99.07	99.25	95.34	99.06	**100.00**	99.82	**100.00**	**100.00**
9	25.00	0.00	47.05	66.66	63.15	88.88	94.44	87.50	**100.00**
10	75.00	72.46	75.75	90.40	90.13	97.14	97.52	**98.20**	97.02
11	83.99	77.23	76.80	95.38	89.81	97.55	98.57	**99.39**	98.95
12	72.85	52.99	65.57	82.95	75.74	95.69	**97.83**	96.21	97.75
13	98.05	95.68	98.35	88.64	98.91	**100.00**	**100.00**	99.39	**100.00**
14	93.44	96.22	93.19	97.89	91.90	**100.00**	99.54	99.80	98.24
15	48.45	54.18	66.97	87.89	89.26	97.39	**98.72**	95.47	96.54
16	78.49	83.33	81.25	**98.80**	**98.80**	85.54	97.83	97.30	**98.80**
OA(%)	80.16	74.18	79.05	92.45	89.00	97.65	98.49	98.18	**98.60**
AA(%)	73.96	64.75	77.08	88.36	86.22	96.12	98.03	96.51	**98.87**
K×100	77.79	70.32	76.02	91.42	87.48	97.31	98.27	97.93	**98.40**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:52:32 UTC from IEEE Xplore. Restrictions apply.

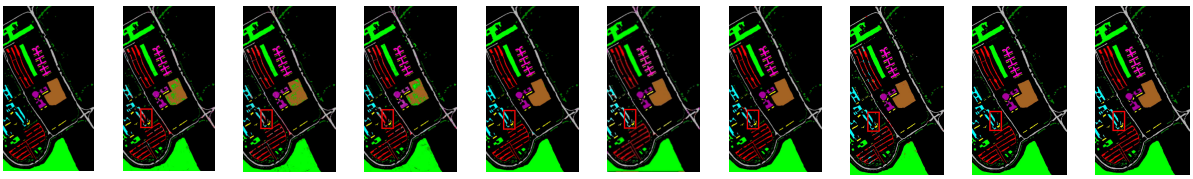
TABLE VI. CLASSIFICATION RESULTS OF THE VARIOUS METHODS FOR THE PAVIA UNIVERSITY DATASET

Class no.	SVM[3]	RNN[5]	1D-CNN[11]	ResNet-50[12]	SF[7]	SSFTT[10]	CS2DT[8]	GSC-ViT[9]	Proposed
1	93.03	92.96	90.23	98.68	94.29	99.97	98.55	99.89	**100.00**
2	98.77	91.15	95.61	99.95	99.27	99.98	99.63	**99.99**	99.95
3	78.32	80.94	76.02	98.73	89.64	99.31	91.77	99.17	**99.89**
4	92.66	93.29	82.04	99.53	99.20	99.35	99.45	**99.92**	99.60
5	99.70	99.34	99.58	**100.00**	**100.00**	99.92	99.81	**100.00**	**100.00**
6	87.19	87.80	87.56	99.78	99.91	**100.00**	97.95	99.98	**100.00**
7	87.44	83.29	71.72	95.66	97.56	**100.00**	99.33	99.15	**100.00**
8	90.98	81.99	87.38	98.55	98.25	99.51	96.85	98.81	**99.57**
9	99.52	99.61	99.46	99.65	**100.00**	98.35	97.17	99.34	99.88
OA(%)	94.11	92.72	90.66	99.38	98.00	99.83	98.49	99.78	**99.90**
AA(%)	92.01	90.71	87.77	98.95	97.57	99.60	97.83	99.58	**99.87**
K×100	92.15	90.32	87.57	99.18	97.36	99.77	98.00	99.71	**99.87**



GT map SVM RNN 1D-CNN ResNet-50 SF SSFTT CS2DT GSC-ViT Proposed

Figure 2. Comparative classification results on the Indian Pines dataset across different methods.



GT-map SVM RNN 1D-CNN ResNet-50 SF SSFTT CS2DT GSC-ViT Proposed

Figure 3. Comparative classification results on the Pavia University dataset across different methods.

IV. CONCLUSION

The transformer model has been proven to utilize spectral correlations, demonstrating superior classification capabilities in the field of hyperspectral image classification. Nevertheless, the standard self-attention mechanism inherently constrains the model's ability to address class imbalance in HSI data, due to limitations in both its network architecture and training methodology. To solve the above-mentioned problems, B^{2}SFormer is proposed, where the original self-attention mechanism in the transformer model is extended to sparse self-attention, to help the network focus on the key channel features and reduce the requirement for a large number of samples in training. The two-branch network structure is then designed with an extra reverse sampling branch, to better utilize the samples in the tail classes in training. In order to verify the superiority of the proposed model, eight representative methods were chosen and compared. The final results indicated that the proposed B^{2}SFormer framework shows a certain improvement in solving the category imbalance problem in the field of hyperspectral remote sensing image classification.

ACKNOWLEDGMENT

This work was partially supported by the National Natural Science Foundation of China (Grant 42001307) and the Natural Science Foundation of Ningxia (Grant 2024AAC05029).

REFERENCES

[1] Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep Learning for Hyperspectral Image Classification: An Overview," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 9,pp. 6690-6709, 2019.

[2] G. Camps-Valls, D. Tuia, L. Bruzzone and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE signal processing magazine*, vol. 31, no. 1, pp. 45-54, 2013.

[3] F. Melgani, and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 42, no. 8, pp. 1778-1790, 2004.

[4] H. Lee, and H. Kwon, "Going Deeper With Contextual CNN for Hyperspectral Image Classification," *IEEE Transactions on Image Processing*, vol. 26, no. 10, pp. 4843-4855, Oct. 2017.

[5] Z. Zhong, J. Li, D. A. Clausi, and A. Wong, "Generative Adversarial Networks and Conditional Random Fields for Hyperspectral Image Classification," *IEEE Transactions on Cybernetics*, vol. 50, no. 7, pp. 3318-3329, 2020.

[6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proceedings of the Neural Information Processing Systems*, 2017, pp. 6000-6010.

[7] D. Hong, Z. Han, J. Yao, L. Gao, B. Zhang, A. Plaza, and J. Chanussot, "SpectralFormer: Rethinking Hyperspectral Image Classification With Transformers," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-15, 2022.

[8] H. Xu, Z. Zeng, W. Yao, and J. Lu, "CS2DT: Cross Spatial-Spectral Dense Transformer for Hyperspectral Image Classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 20, pp. 1-5, 2023.

[9] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral Image Classification Using Groupwise Separable Convolutional Vision

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:52:32 UTC from IEEE Xplore. Restrictions apply.

Transformer Network," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1-17, 2024.

[10] L. Sun, G. Zhao, Y. Zheng and Z. Wu, "Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image Classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-14, 2022.

[11] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *Journal of Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770‑778.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:52:32 UTC from IEEE Xplore. Restrictions apply.